EKITI STATE UNIVERSITY ANTHEM

Verse 1

Hail! Hail!! Ekiti State University,
This lofty vision, which our fathers bore
Blessed be thy diligence
Mustered by great dedication and passion
For Knowledge, Honour and Service.

Chorus

EKSU! EKSU!! EKSU!!!
Shine in all spheres,
I say shine,
The goal is clear,



To birth the best brains,
For strength that builds our Nation.

Verse 2

Hail! Hail!! Ekiti State University,
The fountain from which we all draw with joy,
We acknowledge God and His Grace,
For our institution fostered with learning and culture,
Great may you remain forever and ever

Chorus

EKSU! EKSU!! EKSU!!!
Shine in all spheres,
I say shine,
The goal is clear,
To birth the best brains,
For strength that builds our Nation.

CGPA CALCULATION @EASE of 5.00 Grading System.

GRADE
5 = A
4 = B
3 = C
2 = D
1 = E
0 = F
.....
SCORE
1. 70-100 = A
2. 60-69 = B
3. 50-59 = C
4. 45-49 = D
5. 40-44 = E
6. 0-39 = F
.....
So:
1. 4.50 - 5.00 = First Class
2. 3.50 - 4.49 = 2nd Class Upper
3. 2.40 - 3.49 = 2nd Class Lower
4. 1.50 - 2.39 = 3rd Class
5. 1.0 - 1.49 = Pass
.....
WHAT IS CGPA & GPA??
CGPA - Cumulative Grade Point Average. (This is what you have after both 1st and 2nd semester .... In summation and Division, amounting to it)
GPA - Grade Point Average (This is what you have after every semester)
C.U - Credit Units
TCU - Total Credit Unit
G.P - Grade Point
TGP or TQP - Total Grade Point or Quality Point

Thus;
GPA ..... This is derived by dividing the Total Grade/Quality point (TQP/TGP) by the Total Credit unit (TCU) for that semester.
For example:
If in a semester, a student earned 90 quality points for 21 credit units, the GPA will be;
GPA = TGP/TCU
90/21
= 4.28

HOW TO CALCULATE CGPA:
CGPA is derived by adding the Total Quality Points (TQP) for all the semesters to date and dividing by the Total Credit Units to date for all semester... 1st & 2nd.
For example:
If in a semester a student earns a QP of 90 for 21 CU in the first semester and earns a QP of 87 for 20 CU in the second semester, his/her CGPA will be;
CGPA = TQP/TCU
= (90+87) / (21+20)
= 177 / 41
= 4.31

Your CGPA for that session will be, CGPA = 4.31
..... Breaking it Down:
√ Let's Consider a 1st Semester Result as shown below;

S/n	Units.	Grade	G.P
1.	3	× (A) 5	= 15
2.	2	× (B) 4	= 8
3.	2	× (A) 5	= 10
4.	3	× (A) 5	= 15
5.	3	× (C) 3	= 9
6.	3	× (D) 2	= 6
7.	2	× (E) 1.	= 2
	TCU=18		TGP= 65

GPA = TGP/TCU
= 65/18
= 3.61

√√ Let us now consider 2nd Semester Result.
NB: We'd carry out same procedure in 1st semester then apply here.
But let's assume our TCU & TGP to be = 20 and = 93 respectively.
GPA for second semester GPA = TGP/TCU
93/20
= 4.65

.....
√√√ You'll add first semester and second semester together to get CGPA. AS As shown below;
Mathematically;
CGPA = TGP+TGP(1st & 2nd semester) / TCU+TCU(1st & 2nd Semester)
CGPA = 65+94/ 18+20
= 159 / 38
= 4.184210







NB: **Always approximate your point to 2 decimal places.**
Thus: Your main CGPA = 4.18

STUDENTS' UNION
EKITI STATE UNIVERSITY

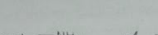
COSMAS

AT SUG TOP SEAT

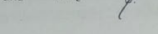
SUPPORT, PRAY, CANVASS!

STUDENTS' UNION
COSMAS
AT SUG TOP SEAT
SUPPORT, PRAY, CANVASS!

functional groups present in 2-Hydroxypropanal is:

A  and 

B  and 

C.  and 

D.  and 

17. The IUPAC name of $(CH_3)_2C=CHCH_3$ is : (A). 2-Methylpent-2-ene (B). 3-Methylpent-3-ene (C). 2-Methylbut-2-ene (D). 2-Methylbut-3-ene

18. The empirical formula of ethanoic acid is: (A). C_2H_2O (B). CH_2O (C). CH_3COOH (D). $HCOOH$

19. A mixture of sugar and talc powder can be separated by: (A). sublimation and dissolution (B). dissolution and evaporation (C). distillation and evaporation (D). evaporation and crystallization.

20. The molecular formula of glucose is (A). $C_6H_{12}O_6$ (B). $C_6H_{10}O_5$ (C). $C_6H_8O_2$ (D). $C_6H_6O_6$

21. An element with atomic number 12 is likely to have the same chemical properties as those elements in

(A). Group I (B). Group II^A (C). Group II^A (D). Group IV^A

22. The valence electrons of Ca are contained in …… (A). 2s orbital (B). 3s orbital (C). 3p orbital (D). 4s orbital.

23. Out of the following arrangements is in order of decreasing electropositivity. (A). F, B, Be, N and Li (B). Li, Be, B, N and F (C). F, N, B, be, and Li (D). Li, N, B, F, and Be.

24. The energy required to remove one or two electron(s) from a 4s orbital of gaseous potassium and calcium atoms respectively is…… (A). Ionization energy (B). Kinetic Energy (C). Electron affinity (D). Electronegativity.

25. Which of the following electronic configuration indicates an atom with the highest ionization energy? (A). $1S^2 2S^2 2P^6$ (B). $1S^2 2S^2 2P^6 3S^2 3P^6 4S^1$ (C). $1S^2 2S^2 2P^6 3S^2 3P^6 4S^2$ (D). $1S^2 2S^2 2P^6 3S^2 3P^6 3D^{10} 4S^2$

26. An element that has atomic number 18 and atomic mass 40 belongs to group, period and block… … (A). Group 2, Period 3, and S-block (B). Group 4, Period 2, and P-block (C). Group 8, Period 3, and P-bloc. (D). Group 6, Period 3, and D-block .

27. The electronic configuration of S^{2-} may be represented by….(A). $1S^2 2S^2 2P^6 3S^2 2P^4$ (B). $1S^2 2S^2 2P^6 3S^2 3P^6$ (C). $1S^2 2S^2 2P^6 3S^2$ (D). $1S^2 2S^2 2P^6 3S^2 3P^4$.

28. Aufbau principle states that orbitals of the lowest energy level have to be occupied before the orbitalsof the highest level. Which of these arrangements agree with the principle?. (A). 1s 2s 3s 4s 5s (B). 1s 2s 2p 3s 4s (C). 1s 2s 2p 3s 3p 4s 3d (D). 1s 2s 2p 3s 3p 3d 4s.

29. The last two rows of the element in the periodic table represented the lanthanides and the actinide series. These groups are collectively called…… (A). d-block (B). f-block (C). s-block (D). p-block

30. Which of the following group of elements is isoelectronic? (A). O^{2-}, F, Ne, Na^+, Mg^{2+} (B). F, O_2, Ne, Mg^{2+} (C). F, N, Mg, Ne, Na^+ (D). O^{2-}, Ne, Na, Mg, Al^{3+}.

31. The purity of a solid sample can best be determined by its : (A). Boiling point (B). Conductivity (C). Melting point (D). Density

32. An isomer of C_5H_{12} is (A). 2-ethylbutane (B). 2-methylbutane (C). 2-methylpropane (D). 2-propylbutane.



EKITI STATE UNIVERSITY, ADO -EK
FACULTY OF SCIENCE
DEPARTMENT OF CHEMISTRY
SECOND SEMESTER B.Sc. EXAMINATIONS
2018/2019 SESSION
NOVEMBER/DECEMBER, 2019

Course Code: CHM 102 (Introductory Chemistry II) **Units** (4) **Time** 1 ½ Hrs **Question Type:** A

Instruction: Answer all questions and shade the appropriate box

MATRIC /REG NO: ..... DEPT: ..... FACULTY: .....

1	A	B	C	D	21	A	B	C	D	41	A	B	C	D	61	A	B	C	D
2	A	B	C	D	22	A	B	C	D	42	A	B	C	D	62	A	B	C	D
3	A	B	C	D	23	A	B	C	D	43	A	B	C	D	63	A	B	C	D
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20	A	B	C	D	40	A	B	C	D	60	A	B	C	D					



1. The major difference between carbon and other group IVA elements include the following except
(A). Carbon has a smaller size that the rest members (B). Carbon has high electronegativity value and no d-orbital electron (C). Carbon is limited to a coordination number of four (D). Carbon has no unique ability to form multiple bonds
2. An atom X has atomic number of 56 and atomic mass of 135. The nucleus of X then has
(A). 56 protons and 135 electrons (B). 79 protons and 56 electrons (C). 79 protons and 56 neutrons
(D). None of these
3. Each of X' and Y' has a total of 18 electrons. The nuclei of the neutral atoms of X and Y respectively contain
(A). 20 and 20 protons (B) 19 and 19 protons (C) 17 and 19 protons (D). 19 and 21 protons
4. Pauli exclusion principle deals with (A). collision theory of reaction rates (B) electronegativity values of elements (C). quantum numbers of orbitals/electrons (D). reversibility of equilibrium reactions
5. The atomic radius of Be, Mg and Ca are 1.13°A, 1.60°A and 1.97°A respectively. This gradation in atomic radius is due to the fact that (A). electronegativity decrease from Be to Mg to Ca (B). metallicity decreases from Be to Mg to Ca (C). the elements are all in the same period (D). the number of electronic shells increases from Be to Mg to Ca
6. Which of the following arrangements is in the order of decreasing electropositivity
(A). Fluorine, Boron, Berylium, Nitrogen, Lithium (B). Lithium, Berylium, Boron, Nitrogen, Fluorine
(C). Fluorine, Nitrogen, Boron, Berylium, Lithium (D). Lithium, Nitrogen, Boron, Fluorine, Berylium

7 All the following properties decrease down a group in the periodic table except
(A). electron affinity (B). atomic radius (C). ionization energy (D). electronegativity

8 Elements in the same period of the periodic table show
(A). similarities in chemical properties (B). progressive increase in atomic number
(C). progressive decrease in electronegativity (D). similarities in electronic structures

9 The lower the electronegativity difference between two elements, the
(A). higher the polarity of their interatomic bond (B). lower the polarity of their interatomic bond
(C). weaker their interatomic bond (D). higher the possibility of forming a molecule

10. Which of the following halogens has the lowest electropositivity (A). Cl (B). I (C). F (D). Br

11. If 12 and 9 are the atomic numbers of elements A and B respectively, the interatomic bond between them is.....
(A). Dative (B). Covalent (C). Ionic (D) Metallic

12. All the following statements are false about NaCl in the solid state except...
(A). It exists as aggregates of ions. (B). It conducts electric current
(C). It exists as discreet molecules (D). Its ions are linked by metallic bonds.

13. When the Lithium atom forms its ion,
(A). It's atomic number decreases (B). It attains the electronic configuration of helium
(C). It gains an extra electron and becomes negatively charged. (D). It loses one proton.

14. If the electronegativity difference between element A and B is small, the interatomic bond between them is...
(A). dative (B). polar covalent (C). ionic (D) non polar covalent

15. A molecule of Nitrogen contain------bond (A). double (B). triple (C). single (D). multiple

16. The bonds that hold components of metallic, ionic network and molecular solids together are
(A). sea of electrons, electrovalent forces, intermolecular forces, covalent forces respectively
(B). sea of electrons, covalent forces, electrovalent forces, intermolecular forces respectively
(C). sea of electrons, electrovalent forces, covalent forces, electrovalent forces respectively
(D). sea of electrons, covalent forces, intermolecular forces, electrovalent forces respectively

17. Which of the following classification is incorrect (A). $H_2O_{(s)}$, molecular solid (B). $KF_{(s)}$, Ionic solid
(C). $SiC_{(s)}$, covalent solid (D). $S_{(s)}$, metallic solid

18. Be, Cu, Zn and Au adopt which packing methods? (A) HCP, CCP, HCO, CCO respectively (B). CCP, HCP, HCP, CCP respectively (C). HCP, HCP, CCP, CCP respectively (D). CCP, CCP, HCP, CCP respectively

19. A unit cell that contain lattice points only at the corners is known as what? How many lattice point does such unit cell contain? (A). face centered unit. Contains four lattice points (B). primitive unit cell. Contains one lattice point (C). body centered unit cell. Contains two lattice points (D). side centered unit cell. Contains two lattice points

20. The crystal system is a format by which crystal structures are classified and each crystal system is defined by the relationship between edge lengths a, b and c. which axial relationship is representative for monoclinic, orthorhombic and triclinic systems (A). $a \neq b = c$ (B). $a \neq b \neq c$ (C). $a = b \neq c$ (D). $a = b = c$

21. Given that atomic radius is defined as half the distance between the centres of two neighbouring atoms, what is the atomic radius of BCC crystal structures if the unit cell length is "a" (A). a/2 (B). a/4 (C). $a/4\sqrt{2}$ (D). $a/4\sqrt{3}$

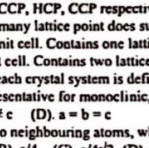
22. Xenon crystallizes in the face-centered cubic lattice and the edge of the unit cell is 620 pm. What is the distance between the nearest neighbours (A). 620 pm (B). 219.23 pm (C). 438.47 pm (D). 525 pm

23. Two ions A^+ and B^- have radii 88 and 200 pm respectively. In the closed packed crystal of compound AB, predict the coordination number of A (A). 4 (B). 8 (C). 6 (D). 12

24. A unit cell of Au has a Au at coordinates 0, 0, 0 and another Au at coordinates 0, ½, ½; ½, 0, ½; ½, ½, 0. How many atoms of Au are there in the unit cell (A). 4 (B) 8 (C). 6 (D). 12

25. Close packing is maximum in the crystal that is —- (A). Simple cubic (B). BCC (C). FCC (D). None of these

26. Solvent extraction is a separation technique which is based on the principle of
(A). Distribution of a solute between two immiscible solvents
(B). Distribution of a solvent between two immiscible solvents
(C). Distribution of a solute between two miscible solvents
(D). Distribution of a solvent between two miscible solutes



27. The purification of liquid systems based on differences in the boiling points of the components is called
(A). Substitution (B). Distillation (C). Distribution (D). Separation

28. Technique based on the distribution of substances between stationary phase and mobile phase is called
(A). Eluent Chromatography (B). Gas Chromatography (C). Column Chromatography (D). Moving